

Experiment Plan Summary

Study Question

Does the *mode* of AI assistance — manual dashboard controls with an advisory AI vs. a chat interface where the AI generates all visualizations — affect data comprehension, confidence calibration, and 1-hour retention?

The Two Conditions

	Group A — Dashboard	Group B — Chat
Interface	Scatter/bar plot with X/Y dropdowns, species checkboxes	Chat input drives all visualization
AI role	Advisory (optional sidebar chat + “Explain plot”)	Generative (controls what gets shown)
User control	High — user chooses what to look at	Low — user asks, AI decides what to show
App	joint_survey - dashboard interface/	joint_survey - chat interface/

Both groups use the same dragon species dataset (3 species, 8 variables) and answer the same 5 in-session survey questions.

Dataset

Fictional dragon species data — 3 types (Forest, Mountain, Sea), 5 numeric variables (claw length/thickness, wingspan, weight, flying speed), plus region and sex. Fictional data removes prior domain knowledge as a confound.

Procedure

1. **Pre-session (5 min)**: Background questionnaire — prior visualization experience, AI chatbot experience, stats comfort (all 1–5 scales). Used as covariates.
2. **Session (20–25 min)**: Explore the interface, answer 5 embedded survey questions with a per-question confidence rating (1–5) after each.
3. **1-hour gap**: Structured classroom activity — no phones, no dashboard access.
4. **Follow-up (15 min)**: Paper-based, no AI. 9 questions across 4 sections.

Randomization: At the session level (not individual) — entire sessions get one condition. Prevents cross-condition contamination from screen-sharing and discussion.

Cover story: Participants are told the study examines “interface design and exploration efficiency” — truthful but does not prime awareness of AI dependency.

In-Session Survey (5 Questions)

Q	Question	Difficulty
Q1	Highest average flying speed by dragon type	Easy — single variable lookup
Q2	Weight vs. flying speed relationship across all types	Medium — cross-variable
Q3	Largest average wingspan by type	Easy — single variable lookup
Q4	Wingspan vs. speed within a single species	Hard — requires filtering first
Q5	Same claw length: Mountain vs. Sea wingspan	Hardest — conditional relationship

Follow-Up Survey (9 Questions, No AI)

Section A — Factual Recall (baseline; both groups expected to perform similarly) - FU-Q1: Which dragon type is heaviest? - FU-Q2: What range describes flying speeds across all types?

Section B — Multi-Variable Reasoning (core test; Group A advantage predicted) - FU-Q3: Within one species, does wingspan predict flying speed? (*retention of Q4*) - FU-Q4: Which

single variable best predicts flying speed? (*not in primary survey — cleanest diagnostic*) - FU-Q5: Same claw length: Mountain vs. Sea wingspan? (*retention of Q5*)

Section C — Transfer (new fictional deep-sea creature dataset presented as a static table) - FU-Q6: Body mass vs. dive speed relationship across 3 species (*mirrors Q2*) - FU-Q7: Open-ended — how would you use a scatter plot to test a within-species relationship? (*process question*)

Section D — Meta-Cognitive - FU-Q8: Open-ended — how did you approach the survey during the session? - FU-Q9: Confidence now vs. during the session?

Each closed question in Sections A–C includes a **confidence rating (1–5)**.

What the Study Can Show

Primary hypothesis: Group B performs comparably in-session (AI gives accurate answers) but Group A outperforms at follow-up — especially on Section B and FU-Q4 — because active exploration builds durable mental models.

Calibration signal: If Group B shows high confidence during the session but lower accuracy at follow-up (when AI is removed), this is a direct measure of autonomy degradation — confidence was anchored to the AI, not to genuine understanding.

FU-Q4 is the key diagnostic: It tests a variable relationship not asked in the primary survey and unlikely to have been generated by Group B’s AI. A Group A advantage in isolation is the strongest evidence that exploration depth matters beyond task performance.

Data Collected

Source	Group A	Group B
Survey responses + per-question confidence	Yes	Yes
Mouse tracking	Yes	Yes
Chat logs	Voluntary AI use	Visualization requests (mandatory) — analyzed separately
Dial-change events (scatter X/Y, species)	Yes — manipulation check	No

Source	Group A	Group B
Group B prompt quality ratings (1–3, two raters)	No	Yes — covariate

Analysis Plan

Power: ~90 participants per group needed to detect a 15 percentage-point difference at $\alpha = 0.05$, 80% power. Small-classroom studies (~40 total) can only detect effects ~25 pp.

Four pre-specified confirmatory tests (Bonferroni-corrected $\alpha = 0.0125$, cluster-robust SEs at session level): 1. Group $\times$ Time interaction on accuracy (in-session vs. follow-up) 2. Section A vs. B vs. C accuracy gradient — expected to steepen for Group B 3. Calibration $\times$ Group interaction at follow-up 4. Group A vs. B difference on FU-Q4 specifically

Subgroup: Group A split into high explorers (3 unique axis combinations) vs. low explorers using dial-change logs. Active-exploration hypothesis applies only to high explorers.

Qualitative: FU-Q7 and FU-Q8 coded by two independent raters (Cohen’s kappa for reliability) for: AI-reference language, self-directed exploration language, and reasoning justifications.

Key Limitations

#	Issue	Severity	Status
L1	Same-room randomization violates SUTVA	Critical	Fixed — session-level randomization
L2	Testing effect confounds FU-Q3 and FU-Q5	Critical	Partially addressed — FU-Q4 is cleaner
L3	No power analysis	Critical	Fixed — added to Section 8
L4	Generation effect overstated for dropdown interaction	Moderate	Reframed as self-directed attention

#	Issue	Severity	Status
L5	Transfer task changes both format and domain simultaneously	Moderate	Partially addressed — label as near-transfer
L6	No manipulation check on Group A exploration	Moderate	Fixed — dial-change event logging
L7	Single global confidence rating in-session	Moderate	Fixed — per-question ratings
L8	Chat volume not comparable across groups	Moderate	Fixed — analyzed separately
L9	1-hour gap may be too short to differentiate groups	Minor	Acknowledged; harder follow-up Qs added
L10	FU-Q4 correct answer unverified	Minor	Action required before data collection
L11	No pre-registration	Minor	Required before first session runs
L12	Group B prompt quality uncontrolled	Minor	Fixed — quality ratings as covariate
L13	Demand characteristics from partial disclosure	Minor	Fixed — cover story added
L14	Incomplete ethics documentation	Minor	Fixed — IRB, consent, withdrawal added
L15	Cluster non-independence in analysis	Minor	Fixed — cluster-robust SEs

Ethics

- IRB approval required before data collection
- Informed consent obtained; participants may withdraw without penalty
- Cover story used; full debrief after follow-up survey
- No PII collected; session tokens only; data retained 5 years then deleted